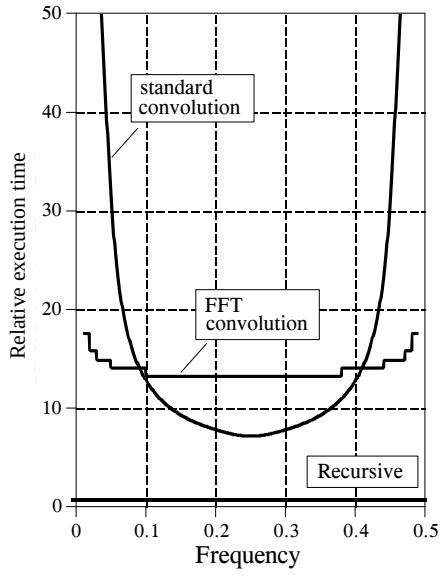

FIGURE 21-5
Comparing FIR and IIR execution speeds. These curves shows the relative execution times for a windowed-sinc filter compared with an equivalent six pole Chebyshev recursive filter. Curves are shown for implementing the FIR filter by both the standard and the FFT convolution algorithms. The windowed-sinc execution time rises at low and high frequencies because the filter kernel must be made longer to keep up with the greater performance of the recursive filter at these frequencies. In general, IIR filters are an order of magnitude faster than FIR filters of comparable performance.

Comparing these filters for speed is like racing a Ferrari against a go-cart. Figure 21-5 shows how much longer the windowed-sinc takes to execute, compared to a six pole recursive filter. Since the recursive filter has a faster roll-off at low and high frequencies, the length of the windowed-sinc kernel must be made *longer* to match the performance (i.e., to keep the comparison fair). This accounts for the increased execution time for the windowed-sinc near frequencies 0 and 0.5. The important point is that FIR filters can be expected to be about an order of magnitude slower than comparable IIR filters (go-cart: 15 mph, Ferrari: 150 mph).

Match #3: Moving Average vs. Single Pole

Our third competition will be a battle of the time domain filters. The first fighter will be a nine point moving average filter. Its opponent for today's match will be a single pole recursive filter using the bidirectional technique. To achieve a comparable frequency response, the single pole filter will use a sample-to-sample decay of $x = 0.70$. The battle begins in Fig. 21-6 where the frequency response of each filter is shown. Neither one is very impressive, but of course, frequency separation isn't what these filters are used for. No points for either side.

Figure 21-7 shows the step responses of the filters. In (a), the moving average step response is a straight line, the most rapid way of moving from one level to another. In (b), the recursive filter's step response is smoother, which may be better for some applications. One point for each side.

These filters are quite equally matched in terms of performance and often the choice between the two is made on personal preference. However, there are